WORKED SOLUTION

STEP I 1993, Question 15

Probability and Statistics • Bayes' Theorem

> **STEP I 1993, Question 15**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/93-s1-q15

The key idea

This is Bayes' theorem dressed up as a story. The trick is to condition on the *true* direction and ask how likely the native's string of answers is in each case. A Biscii always lies, so repeated answers from a Biscii always agree and always point the wrong way. Repetition therefore makes it more likely that you are talking to a Biscii, and that is why asking one more time can reverse your decision.

Solution

The Ascii on their own

An Ascii answers each question truthfully with probability $\frac{3}{4}$, independently of earlier answers. The question has only two possible answers (left or right), so two answers are the same exactly when both are true or both are false:

$$\mathrm{P}(\text{same answer twice}) = \left(\tfrac{3}{4}\right)^2 + \left(\tfrac{1}{4}\right)^2 = \tfrac{9}{16} + \tfrac{1}{16} = \tfrac{10}{16} = \tfrac{5}{8},$$

as required. The "both true" case contributes $\frac{9}{16}$ of this, so by the definition of conditional probability

$$\mathrm{P}(\text{telling the truth} \mid \text{same answer twice}) = \frac{9/16}{10/16} = \frac{9}{10},$$

which is the second result.

Two answers of "Left"

Let L be the event that left is the correct direction and R the event that right is. Let A and B be the events that the native is an Ascii or a Biscii, with $\mathrm{P}(A) = \frac{11}{16}$ and $\mathrm{P}(B) = \frac{5}{16}$, chosen independently of the geography. The captain knows nothing about the direction to begin with, so we take $\mathrm{P}(L) = \mathrm{P}(R) = \frac{1}{2}$. Let E_2 be the event that the native answers "Left" twice.

Ask yourself Why is it easier to work out $\mathrm{P}(E_2 \mid L)$ than $\mathrm{P}(L \mid E_2)$?

Once we know the true direction, each tribe's behaviour is completely described, so the probability of the answers is easy to write down. Bayes' theorem then turns these round.

If left is correct, an Ascii says "Left" twice by telling the truth twice, with probability $\frac{9}{16}$. A Biscii would lie and say "Right", so cannot produce E_2. Hence

$$\mathrm{P}(E_2 \mid L) = \tfrac{11}{16} \cdot \tfrac{9}{16} + \tfrac{5}{16} \cdot 0 = \tfrac{99}{256}.$$

If right is correct, an Ascii says "Left" twice by lying twice, with probability $\frac{1}{16}$, and a Biscii certainly says "Left" twice. Hence

$$\mathrm{P}(E_2 \mid R) = \tfrac{11}{16} \cdot \tfrac{1}{16} + \tfrac{5}{16} \cdot 1 = \tfrac{11+80}{256} = \tfrac{91}{256}.$$

The priors of L and R are equal, so they cancel in Bayes' theorem:

$$\mathrm{P}(L \mid E_2) = \frac{\frac{1}{2} \cdot \frac{99}{256}}{\frac{1}{2} \cdot \frac{99}{256} + \frac{1}{2} \cdot \frac{91}{256}} = \frac{99}{99+91} = \frac{99}{190}.$$

This is greater than $\frac{1}{2}$, so left is the more likely direction and the captain should go left, with probability $\frac{99}{190}$ of being right.

Three answers of "Left"

Let E_3 be the event of three answers of "Left". The same reasoning applies, now with three independent answers from an Ascii:

$$\mathrm{P}(E_3 \mid L) = \tfrac{11}{16}\left(\tfrac{3}{4}\right)^3 + \tfrac{5}{16} \cdot 0 = \tfrac{11}{16} \cdot \tfrac{27}{64} = \tfrac{297}{1024},$$
$$\mathrm{P}(E_3 \mid R) = \tfrac{11}{16}\left(\tfrac{1}{4}\right)^3 + \tfrac{5}{16} \cdot 1 = \tfrac{11}{1024} + \tfrac{320}{1024} = \tfrac{331}{1024}.$$

Since $\mathrm{P}(E_3 \mid R) > \mathrm{P}(E_3 \mid L)$ and the priors are equal, right is now the more likely direction. Precisely,

$$\mathrm{P}(R \mid E_3) = \frac{331}{297+331} = \frac{331}{628} > \frac{1}{2}.$$

So the captain should go *right*, and he is then correct with probability $\frac{331}{628}$.

Ask yourself How can a third "Left" make left *less* likely?

Each extra "Left" multiplies the Ascii's contribution to $\mathrm{P}(E \mid L)$ by $\frac{3}{4}$. In $\mathrm{P}(E \mid R)$ the Ascii term shrinks by a factor of $\frac{1}{4}$, but the Biscii term stays at $\frac{5}{16}$. A long run of identical answers is just what a Biscii produces, so the evidence increasingly says "this is a liar". A truthful Ascii is consistent too, but only with probability $(\frac{3}{4})^n$, which dies away.

Checks

- After a single "Left", the likelihoods are $\frac{11}{16} \cdot \frac{3}{4} = \frac{33}{64}$ (given L) and $\frac{11}{16} \cdot \frac{1}{4} + \frac{5}{16} = \frac{31}{64}$ (given R), so $\mathrm{P}(L \mid \text{Left}) = \frac{33}{64}$. The sequence $\frac{33}{64} \approx 0.516$, $\frac{99}{190} \approx 0.521$, $\frac{297}{628} \approx 0.473$ rises and then falls, which matches the explanation above.
- With many answers, $\mathrm{P}(L \mid E_n) \to 0$, because the Biscii term dominates.
- We checked all four probabilities by exact enumeration with fractions, and by simulating 10^6 conversations: the simulated values were 0.5214 (exact $\frac{99}{190} \approx 0.5211$) and 0.5266 (exact $\frac{331}{628} \approx 0.5271$).

Common slips

- Forgetting to state the assumption $\mathrm{P}(L) = \mathrm{P}(R) = \frac{1}{2}$. Without some prior on the direction the question cannot be answered.
- Using the $\frac{9}{10}$ from the first part on the whole population. It is a statement about an Ascii only; a Biscii also gives the same answer twice, every time.
- Treating the native's three answers as independent *whoever* the native is. They are independent only once we know the tribe (and the true direction), which is why we condition first.

Beyond STEP

The calculation is an example of Bayesian updating: the ratio $\mathrm{P}(E \mid R)/\mathrm{P}(E \mid L)$, called the likelihood ratio or Bayes factor, measures how strongly the evidence favours one hypothesis. Because the tribe is a hidden variable shared by all the answers, the answers are only *conditionally* independent, and evidence that seems to point one way can end up pointing the other. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of conditional probability and of exactly this kind of paradox.

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